Experimental Determination of Moment of Inertia

Alfred Petersson, Jonathan Johansson

Course
Mechanics
Uppsala University

May 8, 2026

Contents

1 Abstract

1.1

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

2.1

Moment of inertia is a physical property of anything which rotates, charachterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate anc measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of mesuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse by adding a large stationary mass to a smaller rotating mass, showing with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

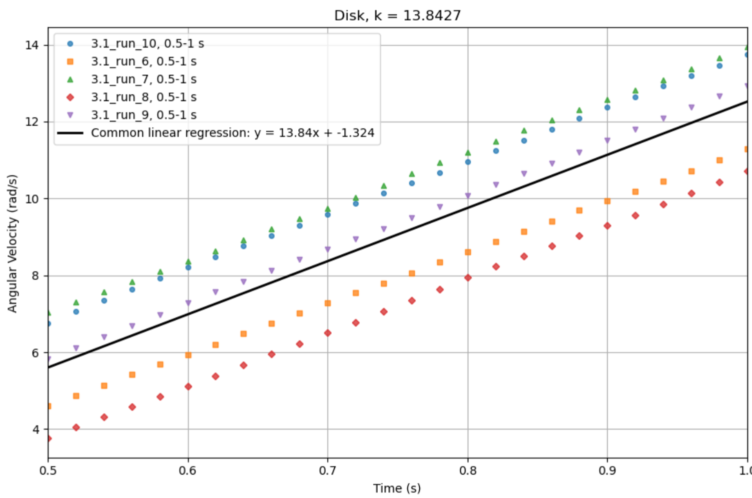

Figure 1: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

3 Theory/Background

3.1

4 Method

4.1

5 Results

5.1

6 Discussion

6.1

7 References

7.1

8 Appendix

8.1

9 Contributions

9.1

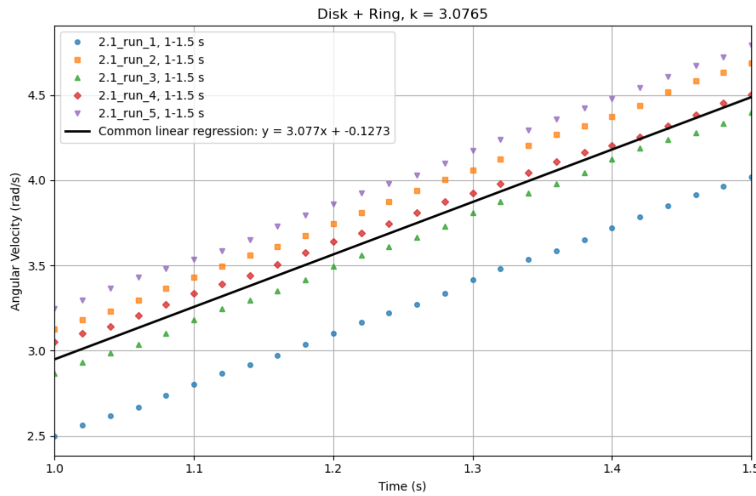

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

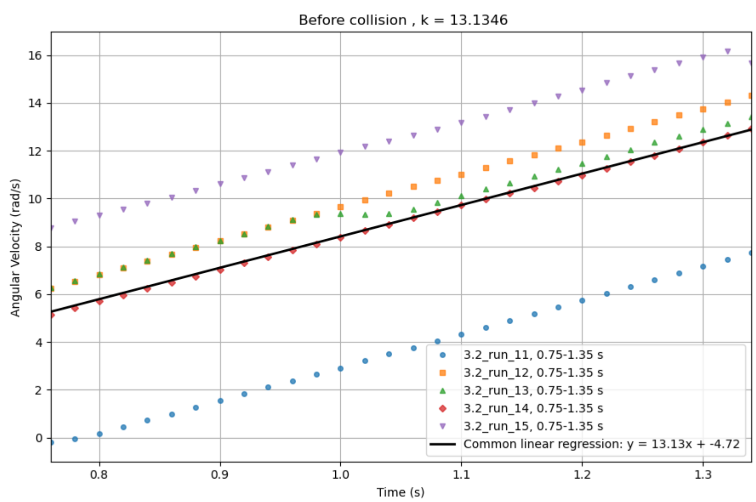

Figure 3: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

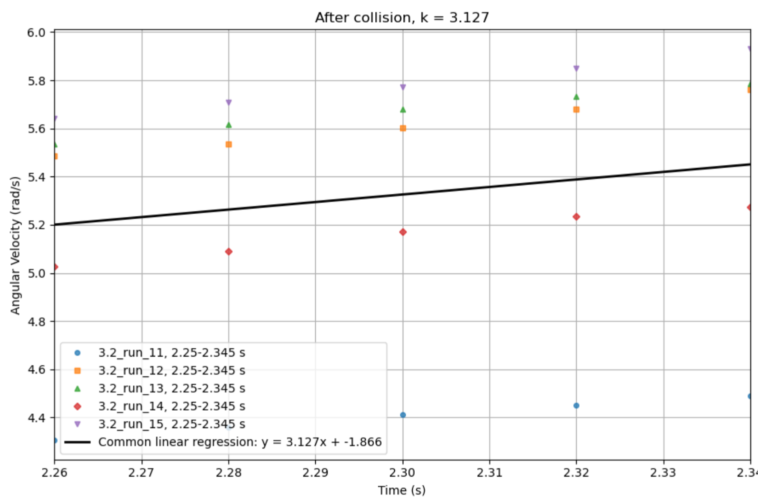

Figure 4: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.